

Feedback by the peer reviewer:

Kyle Gonzales:

Hi Team 17!

~~I'm currently reviewing your draft.~~

Great work on your draft! Retro Games is very well thought out, and your draft was incredibly easy to follow along with.

Does the overview describe what problem is to be solved by a website with DB back end?

Yes, Retro games has experienced massive growth in business and needs a database to help keep up with the increased demand for physical versions of video games.

Does the overview list specific facts?

Yes, Retro Games has a single location in Seattle and has completed over \$250,000 worth of physical video games sales in the past year.

Are at least four entities described and does each entity represent a single idea to be stored as a list?

Yes, the five entities are Employees, Sales, Video_games, Developers, and Customers. Each of these entities represents a single idea that can be stored as a list.

Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints and describe relationships between entities?

The outline of each entity goes into thorough detail. Every attribute is accompanied by a datatype and relevant constraints. Additionally, each entity description describes the relevant relationship, listing the type of relationship and FK's involved.

Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database?

The 1:M relationships are correctly formulated, and the ER diagram clearly illustrates optional vs. required elements in a relationship. There is one M:N relationship between Sales and

Video_games, which is simplified using the Video_game_sales intersection table. The ER diagram clearly presents a logical view of the database.

Is there consistency in a) naming between overview and entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?

Yes, all of the entities have plural names and begin with a capital letter. Every attribute name is entirely lowercase and singular. Any name (entity/table or attribute) with multiple words utilizes an underscore instead of a space.

I think your setup is completely fine as is, but my only suggestion for a small change is in the Video_Games table. The attribute "in_stock" can be derived from "quantity" > 0.

Great work, and good luck with Retro Games!

-Kyle

Jared Yin:

Hi Team 17,

- Does the overview describe what problem is to be solved by a website with DB back end?

Yes, the overview details over the background of the subject area and outlines the specific solution to the issue, which is to provide stability and functionality to the retailer.

- Does the overview list specific facts?

The overview goes over what Retro Games is, what they offer, and their market presence, which is relevant to the issue.

- Are at least four entities described and does each one represent a single idea to be stored as a list?

Entities: Employees, Sales, Video_games, Developers, Customers. 5 total entities and each represents a single ideal.

- Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints and describe relationships between entities?

Every entity established has a corresponding line of text describing its purpose, with the attributes and constraints listed below each.

- Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database?

1:M relationships appears to be correctly established, however lacking at least one M:M relationship.

- Is there consistency in a) naming between overview and entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?

All entity naming schemes are consistently plural and capitalized, attribute naming scheme is consistently singular and lower case.

Ethan Van Hoy:

Hello Group 17,

I enjoyed looking over your project draft!

- **Does the overview describe what problem is to be solved by a website with DB back end?**

Yes, the overview describes how the retailer has had an increase in demand and need a way to track their operations to optimize the business. The website with DB will allow the employer to monitor their employees' sales performance, customers' purchase history, and the popularity of particular games and developers.

- **Does the overview list specific facts?**

Yes, the overview lists specific facts regarding the type of business and products that are sold. It also lists the specific entities that will be tracked on the website and back end DB and the prior year's sales total. Other specific numerical facts that could perhaps be added include the number of employees and the number of games the store usually stocks.

- **Are at least four entities described and does each one represent a single idea to be stored as a list?**

Yes, five unique entities (Employees, Sales, Video_games, and Developers, and Customers) are described, with each representing a single idea related to the business. All the entities can be stored as lists.

- **Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints and describe relationships between entities?**

Each of the five entities have a description that clearly explains the role it plays in the database. They all feature attributes with logical and easy to understand datatypes and constraints. The relationships between entities and their foreign keys are listed for all entities, and they match the ERD diagram.

- **Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database?**

The 1:M relationships between Employees and Sales, Customers and Sales, and Developers and Video_games are all formulated correctly. An employee can be involved in many sales, as can a customer. A developer can produce many games. There is an M:M relationship between Sales and Video_games, with an intersection table between the two labeled on the ERD as Video_game_sales. This relationship makes sense, as a sale can include many video games, and a particular video game title can be purchased in many sales. These relationships are all represented clearly and logically in the ERD, making the database structure easy to follow.

- **Is there consistency in a) naming between overview and entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?**

Yes, there is consistency in the naming of entities and attributes throughout the draft. The entities are plural and utilize underscores rather than spaces, with the first letter of the entity capitalized. All attributes are singular, and utilize snake case with all letters lowercase and underscores between words. These naming conventions are followed in the entity / attributes list as well as the ERD.

Preet Patel:

- **Does the overview describe what problem is to be solved by a website with DB back end?**

Yes it does describe the problem the company experienced a surge in their sales which ended up being 250,000 in sales in the past year and the DB backend will track retro games inventory, employees, customer and sales by implementing this DB the company enhance their operational efficiency and will have accurate inventory control.

- **Does the overview list specific facts?**

Yes, It states that Retro Games is a unique retailer in Seattle area and the company had 250,00 in sales in the past year.

- **Are at least four entities described and does each one represent a single idea to be stored as a list?**

Yes it does have at least five entities.

- Employees
- Sales
- Video_games
- Developers
- Customers

- **Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints and describe relationships between entities?**

Yes the outline of the entity details describes the purpose of each list attribute datatypes and constraints and it does also describe the relationships between entities

- **Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database?**

Yes the 1:M relations correctly formulated and there also a M:M relationship between Sales and Video_games, ERD also presents a logical view of the database

- **Is there consistency in a) naming between overview and entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?**

Yes I did find consistency in the naming, entities and their attributes and the EDR was also easy to follow aswell.

Great work guys!! I hope you guys have a great week!

DRAFT 2 FEEDBACK:

- **Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?**
 - - It looks like your schema presents a physical model that follows your Database outline, as well as the ER diagram that you created. All of the updates you made reflect the corresponding data.
- **Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?**
 - There is a minor inconsistency in the naming scheme used between your schema and the other aspects of your PDF (overview, outline, and ER diagram). Each of the table names are not capitalized, whereas they are everywhere else in your design document. This is just a minor fix, but otherwise, every other aspect of your design is consistent.
- **Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?**
 - The schema is quite easy to read, as there are no relationship lines crossed and the tables are spaced out neatly and cleanly.
- **Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?**

- Intersection tables are properly formed, with the one being "VideoGameSales." There are two FKs (sale_id and video_game_id) being used to facilitate this M:N relationship between the two tables, VideoGames and Sales.
- **Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?**
 - After a quick review, the sample data does not suggest any non-normalized issues (no dependencies or redundancies).
- **Is the SQL file syntactically correct? This can be easily verified by using PhPMysqlAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)**
 - The SQL file was able to be imported flawlessly. No bugs here, all of your SQL syntax was correct. Good job!
- **In the SQL, are the data types appropriate considering the description of the attribute in the database outline?**
 - In the SQL, some of your data types can be updated. For instance, in the Customers table, the zipcode has a type of 'varchar(55).' All zipcodes are digits, so this can be changed to an INT. Additionally, if you are only doing US zip codes, then it might be a good idea to make this int(5), or int(10) (the longest zipcode in the world is in Iran).
- **In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?**
 - In the SQL dump, an appropriate number of CASCADE operations are declared. Additionally, the PK and FKs are correctly defined in the Schema as well. Great work!
- **In the SQL, are relationship tables present when compared to the ERD/Schema?**
 - It appears that the relationship tables are present in the SQL when compared to both the ERD and Schema on your PDF file. Good work!
- **In the SQL, is all example data shown in the PDF INSERTED?**
 - All of the example data shown in the PDF file is also represented within the SQL dump. No issues here, good job!
- **Is the SQL well structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?**
 - The SQL file has header comments for each main query, detailing what each query is doing to the database. This made it readable for me, and definitely will for others not immediately familiar with this database implementation.

Ryan Pak [2d](#)

- **Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?**
 - The schema does follow the outline and ERD. Entities and attributes are represented properly.
- **Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?**

- There is consistent plurality and case capitalization for entities and attributes. In your schema, I'd capitalize the table names. Unsure if camel case is required for the attributes.
- **Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?**
 - The schema is easy to read and does not cross.
- **Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?**
 - There is an intersection table videogamesales.
- **Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?**
 - The sample data does not have any dependencies.
- **Is the SQL file syntactically correct? This can be easily verified by using PhPMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)**
 - The SQL file loaded correctly.
- **In the SQL, are the data types appropriate considering the description of the attribute in the database outline?**
 - I would make the phone numbers varchar so that math operations aren't performed.
- **In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?**
 - The primary and foreign keys are defined and CASCADE is declared properly.
- **In the SQL, are relationship tables present when compared to the ERD/Schema?**
 - The relationship tables are present in SQL when compared to the schema.
- **In the SQL, is all example data shown in the PDF INSERTED?**
 - The inserted SQL data matched that in the PDF.
- **Is the SQL well structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?**
 - Queries are commented describing what they do. Could possibly add description comments if wanted. It appears to be hand authored.

Lucas Stephens [2d](#)

- Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?

There is a small issue with the ERD, where the M:N relationship between VideoGames and Sales does not have crow's foot notation on the sales side.

- Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?

There is some difference in naming between the outline and relationship texts. There is also a difference in capitalization between the schema and ERD.

- Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?

Yes, schema is very easy to read and visually appealing.

- Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?

Yes, the intersection table correctly facilitates the M:N relationship

- Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?

I didn't see any issues with normalization from the samples data.

- Is the SQL file syntactically correct? This can be easily verified by using PhPMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)

Yes, there were no errors when loading the SQL file.

- In the SQL, are the data types appropriate considering the description of the attribute in the database outline?

Yes, all data types are correct. I would recommend putting a limit on the phone number ints so there is a specific length specified.

- In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?

The primary and foreign keys are correctly defined. Cascade operations are declared and they all are appropriate. The cascade of customer_id in sales means records of the sales will be deleted if a customer is deleted which could be bad for keeping a history of sales, but that is up to preference.

- In the SQL, are relationship tables present when compared to the ERD/Schema?

Yes, all relationships are correctly defined.

- In the SQL, is all example data shown in the PDF INSERTED?

Yes, all sample data is inserted into the database.

- Is the SQL well structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?

Yes, SQL is well structured and there are minimal but effective comments.

Overall, great job! There are only a couple small things to fix and make sure is consistent.

Daniel Morden [2d](#)

- Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?
 - the schema reflects the outline and er diagram correctly
- Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?
 - the formatting is consistent and matches with the outline and er diagram
- Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?
 - the schema diagram is organized and relationships between tables are clearly presented
- Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?
 - The intersections table VideoGameSales is properly formed to facilitate M:N relationship between VideoGames and Sales.
- Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?
 - the database design seems to normalize data well, so no issues
- Is the SQL file syntactically correct? This can be easily verified by using PhPMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)
 - the sql file and queries works and can be run
- In the SQL, are the data types appropriate considering the description of the attribute in the database outline?
 - data types used for attributes are reasonable, but you can fix your outline so that date datatype format is YYYY/MM/DD instead of DD/MM/YYYY
- In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?
 - primary and foreign keys are correctly defined, based on the schema. On delete cascade is also appropriately used
- In the SQL, are relationship tables present when compared to the ERD/Schema?
 - the relationship table created in the sql match with the erd and schema
- In the SQL, is all example data shown in the PDF INSERTED?
 - the data inserted in the sql match with the example data pasted in the pdf document
- Is the SQL well structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?
 - the sql is well structured and clear, but should maybe include a bit more explaining in the commenting

Actions based on the feedback:

Based on Kyle's feedback we as a group, decided to remove the "in_stock" variable. We agree that the "quantity" attribute determines if the game is in stock. The TA also mentioned fixing my ERD to represent an ERD rather than a schema. That has been fixed. Everyone else did not mention any other recommendations.

Based on the second draft feedback, we fixed the ERD to make sure Sales table have two crow's feet rather than one. We also fixed some inconsistencies on our naming convention from Video_game_sales to VideoGameSales to match the schema. In addition, we were told that our schema's tables are not capitalized. After troubleshooting, we were not able to fix this issue. We did not notice any other recommendations.

Upgrades to the Draft version:

We have decided to change a few attributes. For the Sales table, we changed the attribute of "total_sold" to "sale_revenue". We feel that it would make more sense to track the total revenue of videogames sold. In addition, we added "list_price" to the intersection table called VideoGameSales. The reason for this is because the videogame will have the original price of said game and when it sells it will sell to whatever the list_price is at the time. We feel this makes sense for the two entities. After the second draft we decided to remove genre from VideoGames, we removed it as it wasn't really important information for the table.

Retro Games

By Jerome Cagado and Gabriel-Jean Bertrand

Overview

Retro Games is a unique gaming retailer exclusively offering physical versions of video games in the Seattle area. The business has experienced a surge in sales, reaching

\$250,000 in the past year. In response to the video game market now favoring digital copies over physical copies, Retro Games needs assistance in a database system to keep up with the growing demand.

The database-driven website will track Retro Games inventory, employees, customers, and sales. By implementing this database, Retro Games will enhance their operational efficiency, ensuring accurate inventory control.

Database Outline

Employees: records the details of Employees who work for Retro Games.

- **employee_id:** INT, auto_increment, unique, not NULL, PK
- **employee_fname:** VARCHAR(45), not NULL
- **employee_lname:** VARCHAR(45), not NULL
- **employee_phone:** INT, not NULL
- **hire_date:** DATE, YYYY/MM/DD
- Relationship:
 - A 1:M relationship between **Employees** and **Sales** with **employee_id** as an FK in **Sales**.

Sales: records details of video games sold.

- **sale_id:** INT, auto_increment, unique, not NULL, PK
- **employee_id:** INT, not NULL, FK
- **customer_id:** INT, not NULL, FK
- **sale_revenue:** DECIMAL, not NULL
- **sold_date:** DATE, YYYY/MM/DD
- Relationship:
 - A M: N relationship between **VideoGames** and **Sales**. Utilizes an intersection table called **VideoGameSales** with **sale_id** as FK.

VideoGames: records details of video games in Retro Games inventory.

- **video_game_id:** INT, auto_increment, not NULL, PK
- **developer_id:** INT, not NULL FK
- **video_game_name:** VARCHAR(45), not NULL
- **price:** DECIMAL, not NULL
- **quantity:** INT, not NULL
- Relationship:
 - A M:N relationship between **VideoGames** and **Sales**. Utilizes an intersection table called **VideoGameSales** with **video_game_id** as FK.

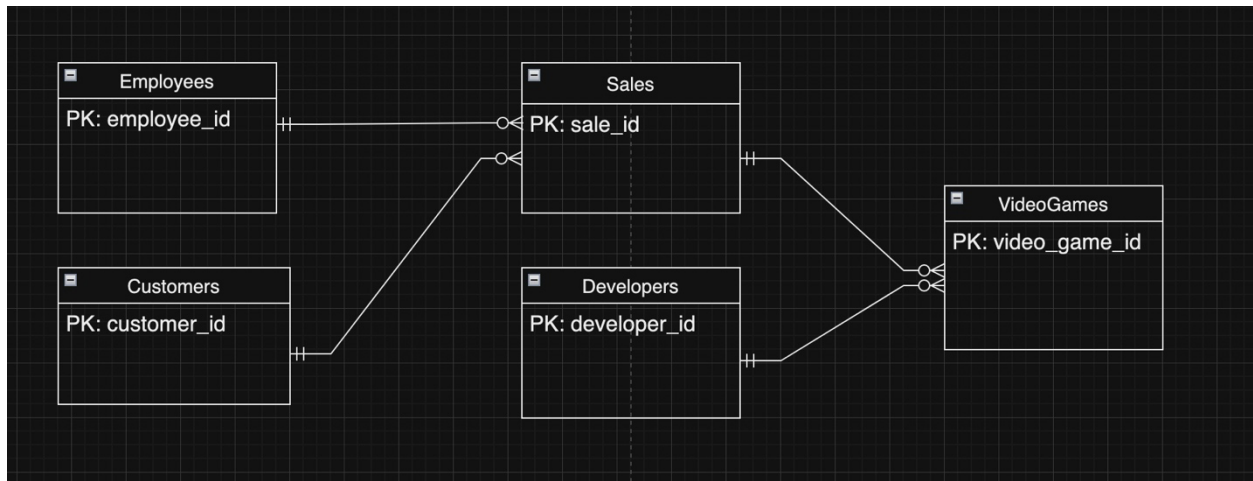
Developers: records details of video game developers.

- **developer_id:** INT, auto_increment, not NULL PK
- **developer_name:** VARCHAR(45), not NULL
- **developer_country:** VARCHAR(45), not NULL
- **developer_email:** VARCHAR(45), not NULL
- Relationship:
 - A 1:M relationship between **Developers** and **VideoGames** with **developer_id** as FK.

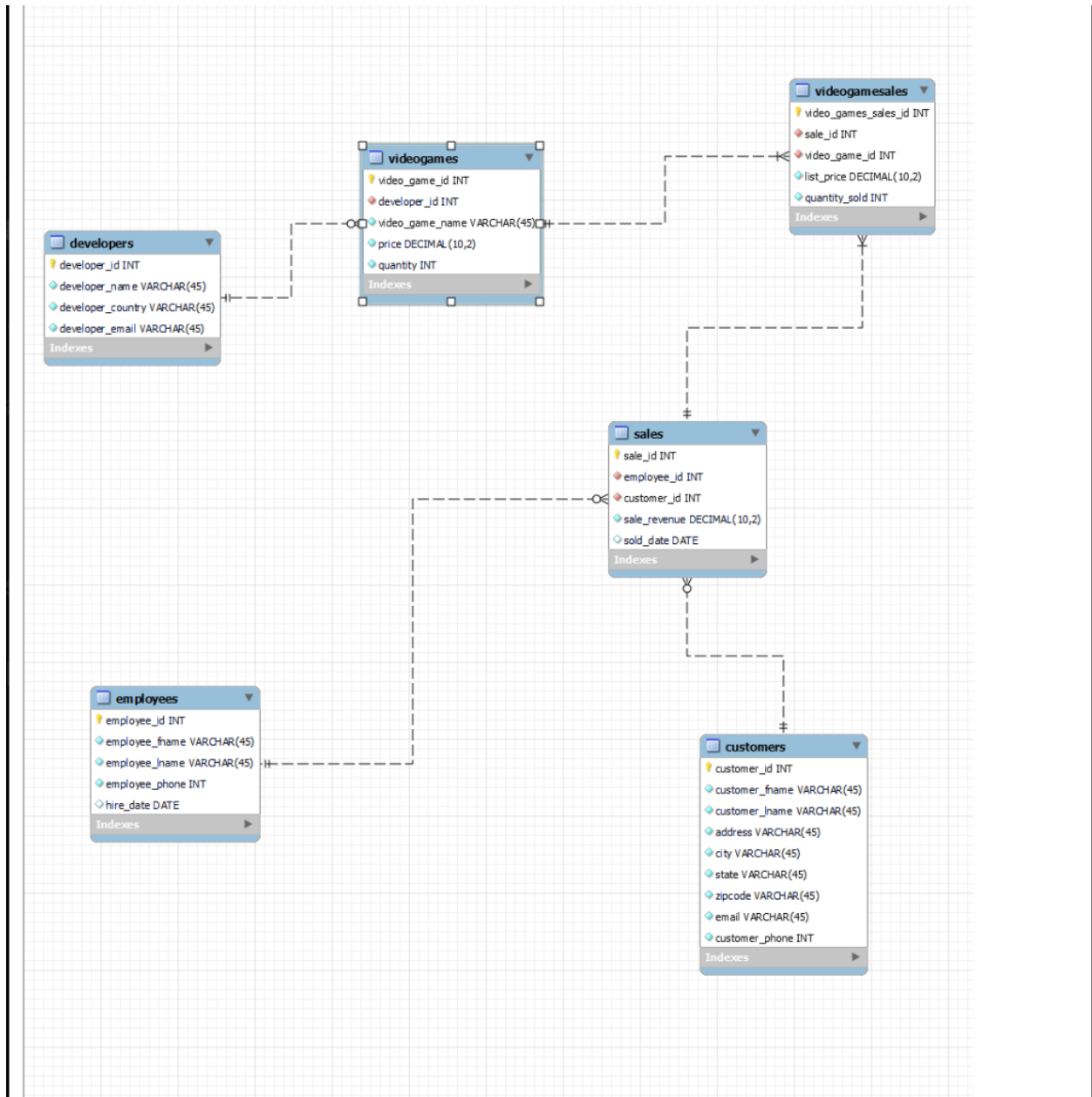
Customers: records details of customers of Retro Games.

- **customer_id:** INT, auto_increment, not NULL, PK
- **customer_fname:** VARCHAR(45), not NULL
- **customer_lname:** VARCHAR(45), not NULL
- **address:** VARCHAR(45), not NULL
- **city:** VARCHAR(45), not NULL
- **state:** VARCHAR(45), not NULL
- **zipcode:** VARCHAR(45), not NULL
- **email:** VARCHAR(45), not NULL
- **customer_phone:** INT, not NULL
- Relationship:
 - A 1:M relationship between **Customers** and **Sales** with **customer_id** as a FK in **Sales**.

Entity-Relationship Diagram



SCHEMA



SAMPLE DATA

Customers

Result Grid									
		Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:	
	customer_id	customer_fname	customer_lname	address	city	state	zipcode	email	customer_phone
▶	1	Sam	Romanano	123 Cool Street	Los Angeles	California	72839	SR@yahoo.com	18727826
	2	Sophia	Rodriguez	321 Nice Avenue	Seattle	Washington	98101	sophia.rodri@gmail.com	14598547
	3	Jonathan	Thompson	432 73rd Street	Minneapolis	Minnesota	55633	J.Teach@gmail.com	18209483
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Developers

Result Grid				
		Filter Rows:		Edit:
				Export/Import:
	developer_id	developer_name	developer_country	developer_email
▶	1	Cool Games	Canada	Cool@gmail.com
	2	Nice Studios	Sweden	Nice@yahoo.com
	3	Ok Company	United States	Okgames@gmail.com
•	NULL	NULL	NULL	NULL

Employees

	employee_id	employee_fname	employee_lname	employee_phone	hire_date
▶	1	John	Smith	14321683	2021-04-25
	2	Terry	Clinton	16548327	2023-05-16
	3	Jill	Cool	15928383	2019-05-03
	4	John	Smith	14321683	2021-04-25
	5	Terry	Clinton	16548327	2023-05-16
	6	Jill	Cool	15928383	2019-05-03
✱	NULL	NULL	NULL	NULL	NULL

Sales

	sale_id	employee_id	customer_id	sale_revenue	sold_date
▶	1	1	1	50.00	2023-07-24
	2	2	3	75.00	2023-07-14
	3	3	2	100.00	2023-10-18
✱	NULL	NULL	NULL	NULL	NULL

VideoGames

	video_game_id	developer_id	video_game_name	price	quantity
▶	1	3	Angry Fish	5.00	10
	2	2	BattleShip	25.00	3
	3	1	Battle Toad	50.00	2
✱	NULL	NULL	NULL	NULL	NULL

VideoGamesSales

	video_games_sales_id	sale_id	video_game_id	list_price	quantity_sold
▶	1	1	1	5.00	10
	2	2	2	25.00	3
	3	3	3	50.00	2
✱	NULL	NULL	NULL	NULL	NULL